

DC-DC module power supply specialized for SiC driver



FEATURES

- High efficiency up to 80%
- SIP package
- I/O isolation test voltage : 3.5kVAC/6kVDC
- Ultra-low isolation capacitance
- Operating ambient temperature range: -40°C ~ +105°C
- Continuous short-circuit protection
- Industry standard pin-out
- IEC60950、UL60950 and EN60950 approved

QA01C is DC-DC module power supply designed for SiC driver requiring two sets of isolation power supply. The mode of mutual connection after two independent outputs is adopted internally for better energy provision of SiC turn-on and turn-off. Output short-circuit protection and self-recovery capabilities are also provided. General application includes:

1. Universal converter
2. AC servo drive system
3. Electric welding machine
4. Uninterruptible power supply (UPS)

Selection Guide

Certification	Part No.	Input Voltage (VDC)	Output		Efficiency at Full Load (%) Min./Typ	Capacitive Load* (μF) Max.
		Nominal (Range)	Voltage (VDC)+Vo/-Vo	Current (mA)+Io/-Io		
UL/CE/CB	QA01C	15 (13.5-16.5)	+20/-4	+100/-100	76/80	220

Note: *The specified maximum capacitive load for positive and negative output is identical.

Input Specifications

Item	Operating Conditions	Min.	Typ.	Max.	Unit
Input Current (full load / no-load)	15V input	--	193/16	--	mA
Surge Voltage (1sec. max.)		-0.7	--	21	VDC
Input Filter		Capacitance filter			
Hot Plug		Unavailable			

Output Specifications

Item	Operating Conditions	Min.	Typ.	Max.	Unit
Voltage Accuracy		See output regulation curve (Fig. 1, Fig. 2)			
Linear Regulation	Input voltage change: ±10%	--	±1.1	±1.3	%/%
Load Regulation	10%-100% load	20VDC output		7	%
		-4VDC output		10	
Ripple & Noise*	20MHz bandwidth	Ripple		60	mVp-p
		Noise		75	
Temperature Coefficient	100% load	--	±0.03	--	%/°C
Output Short Circuit Protection		Continuous, self-recovery			

Note: *Ripple and noise tested with "parallel cable" method, please see DC-DC Converter Application Notes for specific operation methods.

General Specifications

Item	Operating Conditions	Min.	Typ.	Max.	Unit
Isolation Voltage	Input-output Electric Strength test for 1 minute with a leakage current of 1mA max.	3500	--	--	VAC
		6000	--	--	VDC
Isolation Resistance	Input-output insulation at 500VDC	1000	--	--	MΩ

Isolation Capacitance	Input-output capacitance at 100kHz/0.1V	--	3.5	--	pF
Operating Temperature	Derating when operating temperature up to 85°C, (see Fig. 3)	-40	--	105	°C
Storage Temperature		-55	--	125	
Pin Soldering Resistance Temperature	Soldering spot is 1.5mm away from the case, 10 seconds	--	--	300	
Case Temperature Rise	Ta=25°C	--	30	--	
Storage Humidity	Non-condensing	--	--	95	%RH
Switching Frequency	100% load, nominal input voltage	--	95	--	KHz
MTBF	MIL-HDFK-217F@25°C	3500	--	--	K hours

Mechanical Specifications

Case Material	Black plastic; flame-retardant and heat-resistant (UL94 V-0)				
Package Dimensions	19.50 × 9.80 × 12.50mm				
Weight	4.20g (Typ.)				
Cooling Method	Free air convection				

Electromagnetic Compatibility (EMC)

Emissions	Conducted disturbance	CISPR32/EN55032	CLASS B (see Fig. 5 for recommended circuit)		
	Radiated emission	CISPR32/EN55032	CLASS B (see Fig. 5 for recommended circuit)		
Immunity	Electrostatic discharge	IEC/EN61000-4-2	Contact	±6KV	perf. Criteria B

Typical Characteristic Curves

+Vo Output Regulation Curve

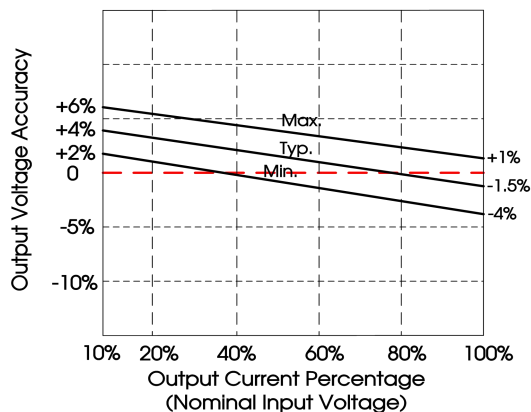


Fig. 1

-Vo Output Regulation Curve

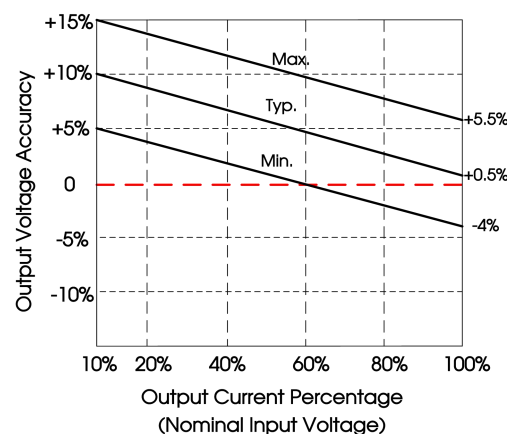


Fig. 2

Temperature Derating Curve

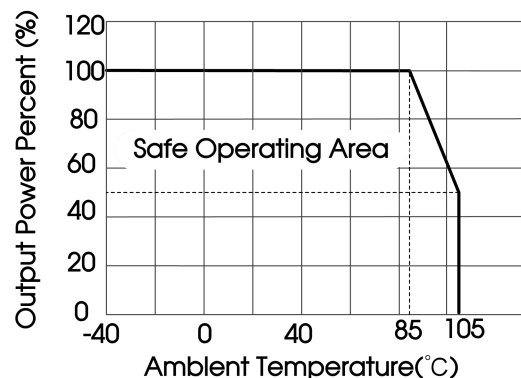
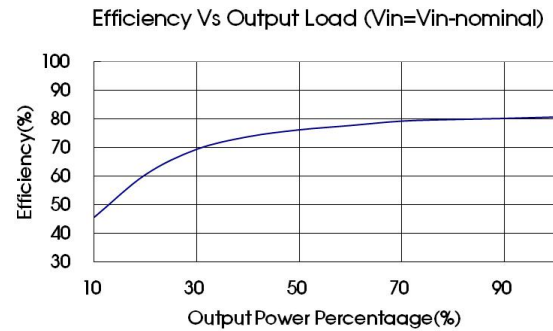
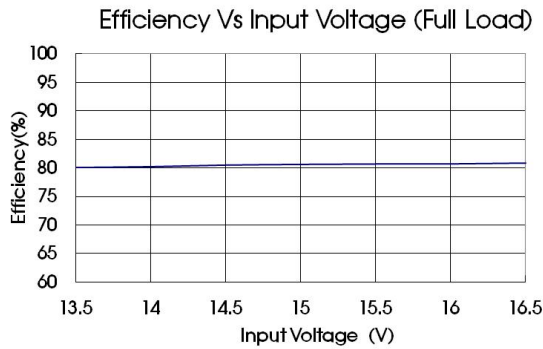


Fig. 3

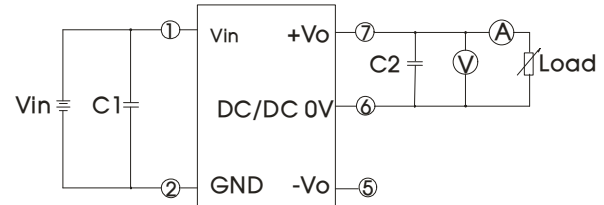
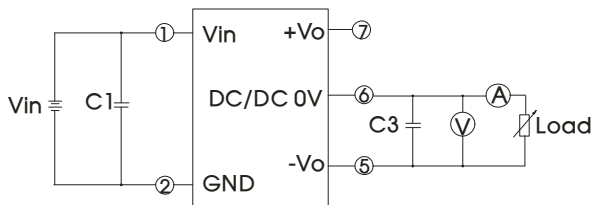


Design Reference

1. Overload Protection

There is no over-load protection under normal operating conditions, we suggest to add an circuit breaker outside in the circuit.

2. Test configurations



Note: C1,C2,C3: 100uF/35V (Low impedance)

3. Typical application

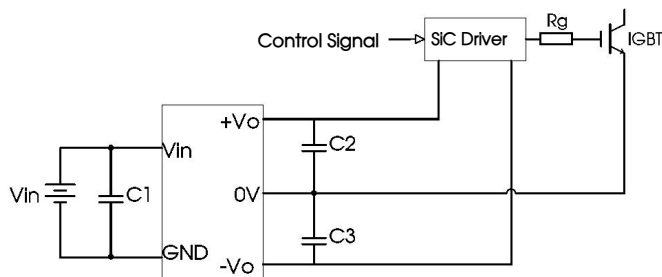


Fig. 4

C1/C2/C3
100uF/35V (Low internal resistance capacitance)

4. EMC typical recommended circuit (CLASS B)

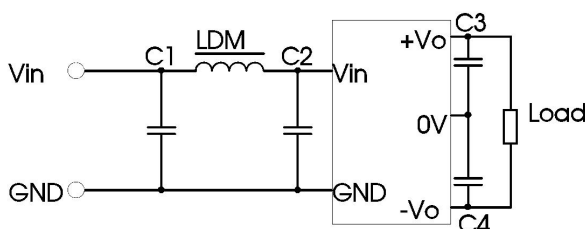


Fig. 5

Input voltage (VDC)		15
EMI	C1/C2	4.7μF /50V
	C3/C4	100μF /35V (Low internal resistance capacitance)
	LDM	6.8μH

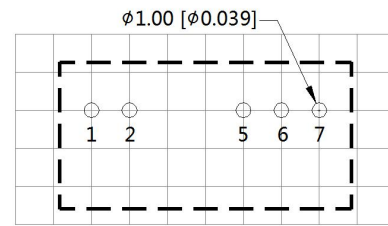
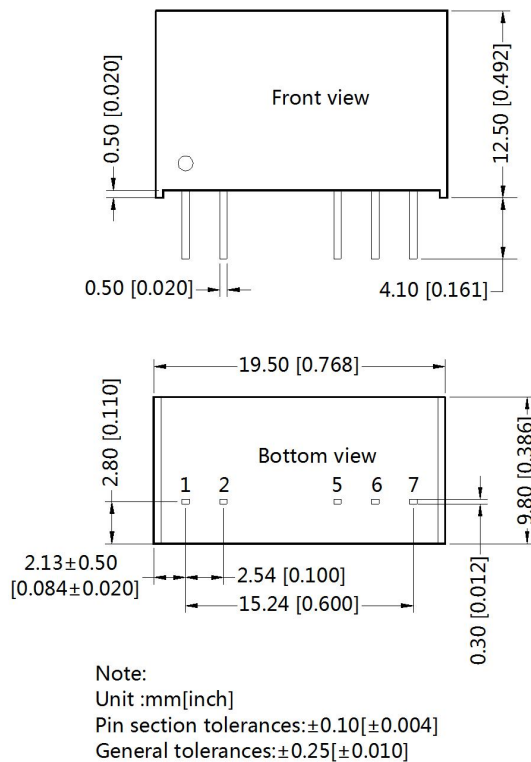
5. The products do not support parallel connection of their output for power expansion purpose or hot-plug.

6. Electrolytic capacitors are recommended for external capacitors at the input or output of the product. Tantalum capacitors are not, otherwise there is a risk of failure.

7. For more information please find the application notes on www.mornsun-power.com

Dimensions and Recommended Layout

THIRD ANGLE PROJECTION 



Pin-Out	
Pin	Function
1	Vin
2	GND
5	-Vo
6	0V
7	+Vo

Notes:

- Packaging information please refer to Product Packaging Information which can be downloaded from www.mornsun-power.com. Packaging bag number: 58200013;
- The lead connecting the power supply module and SiC driver should be as short as possible during use;
- The output filtering capacitor should be as close as possible to the power supply module and SiC driver;
- The peak of the SiC driver gate drive current is high, so low internal resistance electrolytic capacitor is recommended to be used for the power supply module output filter capacitor;
- The average output power of the driver must be lower than that of the power supply module;
- Consider fixing with glue near the module if being used in vibration occasion;
- The max. capacitive load should be tested within the input voltage range and under full load conditions;
- Unless otherwise specified, data in this datasheet should be tested under the conditions of $T_a=25^\circ\text{C}$, humidity<75% when inputting nominal voltage and outputting rated load;
- All index testing methods in this datasheet are based on our company corporate standards;
- The performance indexes of the product models listed in this manual are as above, please directly contact our technicians for specific information;
- We can provide product customization service;
- Products are related to laws and regulations: see "Features" and "EMC".
- Our products shall be classified according to ISO14001 and related environmental laws and regulations, and shall be handled by qualified units

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